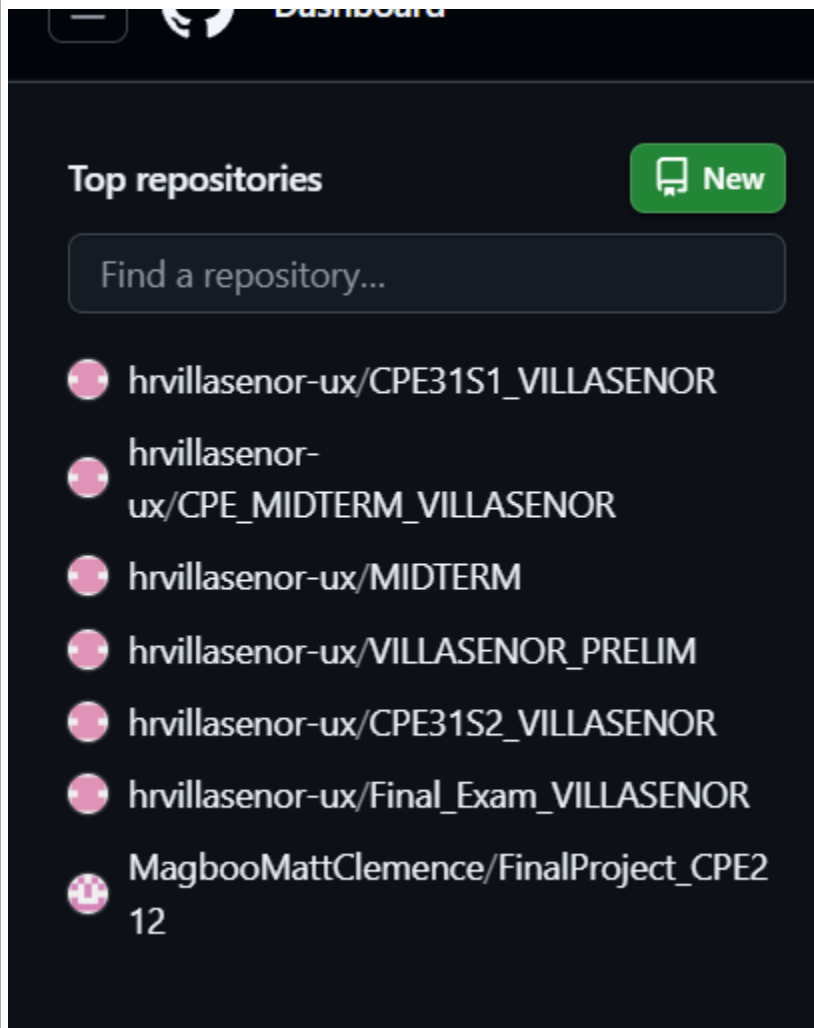


Tools Needed:

1. VM with Ubuntu, CentOS and Ansible installed
2. Web browser

Procedure:

1. Create a repository and label it as "Final_Exam_Surname"



2. Clone your new repository in your VM

```
HANS@LocalMachine: ~/Final_Exam_VILLASENOR
● HANS@LocalMachine:~$ ls
CPE212          Desktop      Final_Exam_VILLASENOR  Pictures  Templates
CPE31S1_VILLASENOR Documents    MIDTERM                Public    Videos
CPE_MIDTERM_VILLASENOR Downloads    Music                  snap      VILLASENOR_PRELIM
○ HANS@LocalMachine:~$
```

3. Create an Ansible playbook that does the following with an input of a config.yaml file and structure inventory file.

3.1 Install and configure one enterprise service that can be installed in Debian and Centos servers

3.2 Install and configure one monitoring tool that can be installed in Debian and Centos servers (if it is a stack there should be option of different host)

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
HANS@LocalMachine:~/Final_Exam_VILLASENOR$ ansible-playbook -i inventory.ini --ask-become-pass con
fig.yaml
BECOME password:

PLAY [all] *****

TASK [Gathering Facts] *****
fatal: [192.168.56.101]: UNREACHABLE! => {"changed": false, "msg": "Failed to connect to the host
via ssh: HANS@192.168.56.101: Permission denied (publickey,password).", "unreachable": true}
fatal: [192.168.56.103]: UNREACHABLE! => {"changed": false, "msg": "Failed to connect to the host
via ssh: ssh: connect to host 192.168.56.103 port 22: No route to host", "unreachable": true}
ok: [192.168.56.104]
ok: [192.168.56.102]

TASK [motd player] *****
ok: [192.168.56.102]
ok: [192.168.56.104]

PLAY [all] *****

TASK [Gathering Facts] *****
ok: [192.168.56.104]
ok: [192.168.56.102]

TASK [prometheus : Install Prometheus on Ubuntu] *****
skipping: [192.168.56.104]
ok: [192.168.56.102]

TASK [prometheus : Create Prometheus user on CentOS] *****
skipping: [192.168.56.102]
ok: [192.168.56.104]
```

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
HANS@LocalMachine:~/Final_Exam_VILLASENOR$ ansible-playbook -i inventory.ini --ask-become-pass con
fig.yaml
ok: [192.168.56.104]

TASK [prometheus : Download Prometheus tarball on CentOS] *****
skipping: [192.168.56.102]
ok: [192.168.56.104]

TASK [prometheus : Extract Prometheus on CentOS] *****
skipping: [192.168.56.102]
ok: [192.168.56.104]

TASK [prometheus : Move Prometheus binary to /usr/local/bin] *****
skipping: [192.168.56.102]
ok: [192.168.56.104]

TASK [prometheus : Create Prometheus config file] *****
skipping: [192.168.56.102]
ok: [192.168.56.104]

TASK [prometheus : Create Prometheus systemd service] *****
skipping: [192.168.56.102]
ok: [192.168.56.104]

TASK [prometheus : Start Prometheus service] *****
skipping: [192.168.56.102]
changed: [192.168.56.104]
```

```
HANS@LocalMachine:~/Final_Exam_VILLASENOR$ ansible-playbook -i inventory.ini --ask-become-pass con
fig.yaml
```

```
TASK [apache : install apache and php for Ubuntu servers] *****
skipping: [192.168.56.104]
ok: [192.168.56.102]
```

```
TASK [apache : install apache and php for CentOS Servers] *****
skipping: [192.168.56.102]
ok: [192.168.56.104]
```

```
TASK [apache : start httpd (CentOS)] *****
skipping: [192.168.56.102]
ok: [192.168.56.104]
```

```
PLAY RECAP *****
192.168.56.101      : ok=0    changed=0    unreachable=1    failed=0    skipped=0    rescue
d=0    ignored=0
192.168.56.102      : ok=5    changed=0    unreachable=0    failed=0    skipped=9    rescue
d=0    ignored=0
192.168.56.103      : ok=0    changed=0    unreachable=1    failed=0    skipped=0    rescue
d=0    ignored=0
192.168.56.104      : ok=12   changed=1    unreachable=0    failed=0    skipped=2    rescue
d=0    ignored=0
```

4.4 Change Motd as "Ansible Managed by <username>"

```
Fig.yaml
● HANS@LocalMachine:~/Final_Exam_VILLASENOR$ ssh HANS@192.168.56.102
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-33-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

9 updates can be applied immediately.
4 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

2 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

*** System restart required ***
Ansible Managed by Hans Villaseñor
Last login: Thu Nov 20 09:33:51 2025 from 192.168.56.101
HANS@Server1:~$ cat /etc/motd
Ansible Managed by Hans Villaseñor
```

```
Connection to 192.168.56.102 closed.
○ HANS@LocalMachine:~/Final_Exam_VILLASENOR$ ssh hans@192.168.56.104
Ansible Managed by Hans Villaseñor
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Thu Nov 20 17:34:24 2025 from 192.168.56.101
[hans@localhost ~]$ cat /etc/motd
Ansible Managed by Hans Villaseñor
[hans@localhost ~]$
```

4. Push and commit your files in GitHub

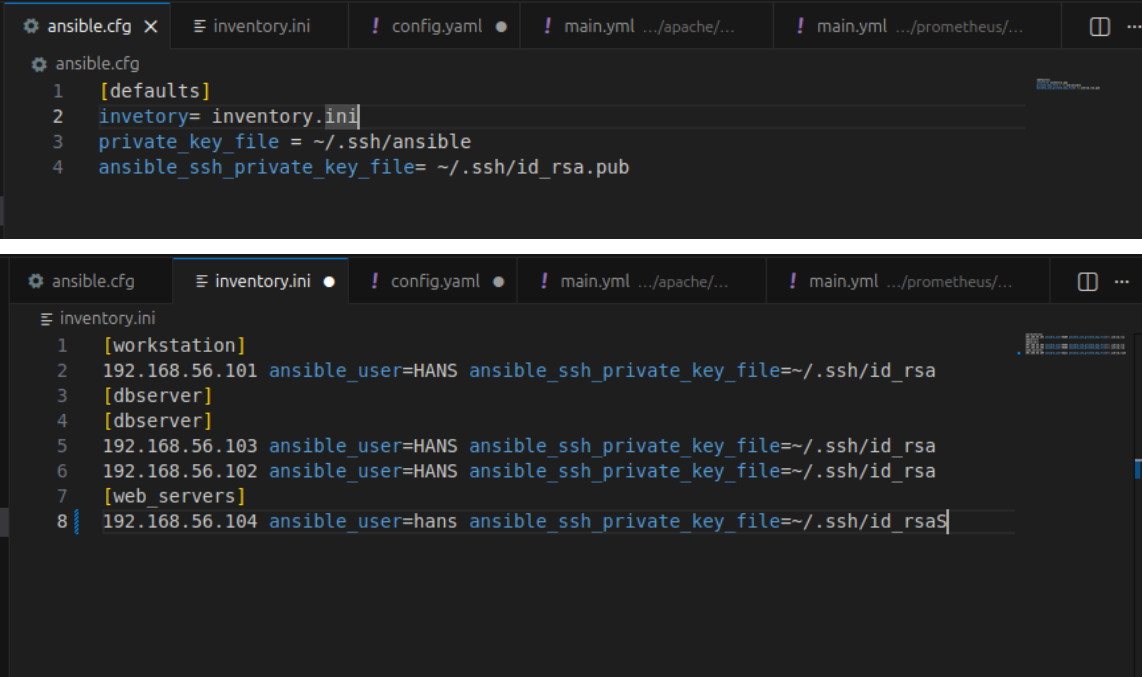


The screenshot shows a GitHub repository named 'FINAL EXAM' by user 'hrvillasenor-ux'. The repository has 1 file (1f5164b · now) and 5 commits. The file list includes:

File	Commit	Time
roles	FINAL EXAM	now
README.md	Initial commit	last week
ansible.cfg	FINAL EXAM	18 hours ago
config.yaml	FINAL EXAM	18 hours ago
inventory.ini	FINAL EXAM	18 hours ago

Below the file list is a 'README' section with the title 'Final_Exam_VILLASENOR'.

5. Make sure to show evidence of input (codes) process (codes successfully running) and output (evidence of installation)



The screenshot shows an IDE with two tabs open: 'ansible.cfg' and 'inventory.ini'. The 'ansible.cfg' tab is active, showing the following code:

```
1 [defaults]
2 inventory= inventory.ini
3 private_key_file = ~/.ssh/ansible
4 ansible_ssh_private_key_file= ~/.ssh/id_rsa.pub
```

The 'inventory.ini' tab is also visible, showing the following code:

```
1 [workstation]
2 192.168.56.101 ansible_user=HANS ansible_ssh_private_key_file=~/.ssh/id_rsa
3 [dbserver]
4 [dbserver]
5 192.168.56.103 ansible_user=HANS ansible_ssh_private_key_file=~/.ssh/id_rsa
6 192.168.56.102 ansible_user=HANS ansible_ssh_private_key_file=~/.ssh/id_rsa
7 [web_servers]
8 192.168.56.104 ansible_user=hans ansible_ssh_private_key_file=~/.ssh/id_rsa$
```

```
ansible.cfg  inventory.ini  ! config.yaml  ! main.yml .../apache/...  ! main.yml .../prometheus/...  ...
! config.yaml
1  ---
2  - hosts: all
3    become: true
4    pre_tasks:
5
6      - name: motd player
7        copy:
8          content: "Ansible Managed by Hans Villasenor\n"
9          dest: /etc/motd
10
11 - hosts: all
12   become: true
13   roles:
14     - prometheus
15     - apache
```

```
ansible.cfg  inventory.ini  ! config.yaml  ! main.yml .../apache/...  ! main.yml .../prometheus/...  ...
roles > apache > tasks > ! main.yml
1  ---
2
3  - name: install apache and php for Ubuntu servers
4    tags: apache2-ubuntu
5    apt:
6      name:
7        - apache2
8        - libapache2-mod-php
9      state: latest
10     update_cache: yes
11     when: ansible_distribution == "Ubuntu"
12
13 - name: install apache and php for CentOS Servers
14   tags: centos-httpd
15   dnf:
16     name:
17       - httpd
18       - php
19     state: latest
20     when: ansible_os_family == "RedHat"
21
22 - name: start httpd (CentOS)
23   tags: centos-httpd
24   service:
25     name: httpd
26     state: started
27     enabled: true
28     when: ansible_os_family == "RedHat"
```

roles > prometheus > tasks > ! main.yml

```
1  ---
2
3  - name: Install Prometheus on Ubuntu
4    tags: prometheus
5    apt:
6      name: prometheus
7      state: latest
8      update_cache: yes
9    when: ansible_distribution == "Ubuntu"
10
11 - name: Create Prometheus user on CentOS
12   tags: prometheus
13   user:
14     name: prometheus
15     shell: /sbin/nologin
16   when: ansible_os_family == "RedHat"
17
18 - name: Download Prometheus tarball on CentOS
19   tags: prometheus
20   get_url:
21     url: https://github.com/prometheus/prometheus/releases/download/v2.45.0/prometheus-2.45.0-linux-amd64.tar.gz
22     dest: /tmp/prometheus.tar.gz
23   when: ansible_os_family == "RedHat"
24
25 - name: Extract Prometheus on CentOS
26   tags: prometheus
27   unarchive:
28     src: /tmp/prometheus.tar.gz
29     dest: /opt/
30     remote_src: yes
31   when: ansible_os_family == "RedHat"
32
```

ansible.cfg

inventory.ini

! config.yml

! main.yml .../apache/...

! main.yml .../prometheus/...

roles > prometheus > tasks > ! main.yml

```
32
33 - name: Move Prometheus binary to /usr/local/bin
34   tags: prometheus
35   copy:
36     remote_src: yes
37     src: /opt/prometheus-2.45.0.linux-amd64/prometheus
38     dest: /usr/local/bin/prometheus
39     mode: '0755'
40   when: ansible_os_family == "RedHat"
41
42 - name: Create Prometheus config file
43   tags: prometheus
44   copy:
45     dest: /etc/prometheus.yml
46     content: |
47       global:
48         scrape_interval: 15s
49
50       scrape_configs:
51         - job_name: 'node_exporter'
52           static_configs:
53             - targets: ['192.168.56.101:9100', '192.168.56.104:9100']
54   when: ansible_os_family == "RedHat"
```

```

56 - name: Create Prometheus systemd service
57   tags: prometheus
58   copy:
59     dest: /etc/systemd/system/prometheus.service
60     content: |
61       [Unit]
62       Description=Prometheus Monitoring
63       After=network.target
64
65       [Service]
66       User=prometheus
67       ExecStart=/usr/local/bin/prometheus --config.file=/etc/prometheus.yml
68
69       [Install]
70       WantedBy=multi-user.target
71   when: ansible_os_family == "RedHat"
72
73 - name: Start Prometheus service
74   tags: prometheus
75   systemd:
76     name: prometheus
77     enabled: true
78     state: started
79   when: ansible_os_family == "RedHat"

```

```

Last login: Thu Nov 20 17:34:24 2025 from 192.168.56.101
[hans@localhost ~]$ cat /etc/motd
Ansible Managed by Hans Villaseñor
[hans@localhost ~]$ which prometheus
/usr/local/bin/prometheus
[hans@localhost ~]$

```

```

Last login: Thu Nov 20 17:34:24 2025 from 192.168.56.101
HANS@Server1:~$ which /etc/prometheus
HANS@Server1:~$ dpkg -l | grep prometheus
ii  prometheus                2.45.3+ds-2ubuntu0.3      amd64
    monitoring system and time series database
ii  prometheus-node-exporter  1.7.0-1ubuntu0.3          amd64
    Prometheus exporter for machine metrics
ii  prometheus-node-exporter-collectors 0.0~git20231018.f5c56e7-1 all
    Supplemental textfile collector scripts for Prometheus node_exporter
ii  python3-prometheus-client 0.19.0+ds1-1              all
    Python 3 client for the Prometheus monitoring system
HANS@Server1:~$

```

 VILLASEÑOR (18 hours ago) Ln 8, Col 22 Spaces: 2 UTF-8 LF {} YAML 

5.1 For your final exam to be counted, please paste your repository link as an answer in this exam.

https://github.com/hrvillaseñor-ux/Final_Exam_VILLASEÑOR.git

Note: Extra points if you will implement the said services via containerization.